A Knowledge Graph for Pancreatic Ductal Adenocarcinoma

Biomarker prioritization, repurposing paths and their limits

Mohammed Sharukh · BIOT 6900 Computational Lab 7 · Fall 2026 · github.com/MohammedSharukh7/biot6900 (module7/)

Overview

I built a Neo4j AuraDB knowledge graph for PDAC that joins the outputs of earlier modules: the CPTAC-PDAC multi-omics ranking from Module 2, germline GWAS variants from the Module 5/6 polygenic score (PGS000725), and curated phenotypes (HPO), pathways (Reactome) and drugs. The final graph has **64 nodes** (29 genes, 11 pathways, 11 drugs, 5 phenotypes, 5 variants, 3 diseases) and **77 relationships** (30 ASSOCIATED_WITH, 20 PARTICIPATES_IN, 12 TARGETS, 10 HAS_PHENOTYPE, 5 OCCURS_IN). All 29 gene symbols were resolved to Ensembl gene IDs through the Ensembl REST API, with one ID per symbol and no shared IDs, so each gene is exactly one node; CEL, which is both a PDAC and a chronic-pancreatitis gene, is a single node with two disease edges rather than two nodes. After loading, a Cypher count in AuraDB returned the same totals by label (Gene 29, Pathway 11, Drug 11, Phenotype 5, Variant 5, Disease 3) and 77 relationships, so nothing was dropped or duplicated between Python and Neo4j.

1. Schema and evidence

The schema follows the question the graph has to answer: which drugs act on PDAC biology, directly or through a related disease. Genes are the hub type, and each gene enters for a stated reason, recorded in a group property: the 15 top Module 2 multi-omics hits, five canonical drivers or pharmacology anchors (KRAS, TP53, CDKN2A, SMAD4, EGFR, which rank 36 to 2444 in Module 2 because the score rewards expression change rather than mutation), five germline GWAS genes linked to the Module 5 variants, and four genes that belong only to a related disease. Phenotypes are what connect PDAC to other diseases, so they are what makes the multi-hop repurposing query possible; I chose two related diseases that share different symptoms for different reasons. Cholangiocarcinoma shares jaundice and weight loss because both tumours obstruct the biliary tree, and chronic pancreatitis shares abdominal pain, steatorrhea and diabetes because both destroy exocrine and endocrine pancreas.

Edge type	Meaning	Evidence stored on the edge
ASSOCIATED_WITH	gene → disease	pvalue (somatic mutation-enrichment p, or GWAS lead-SNP p), Module 2 score, source, confidence
HAS_PHENOTYPE	disease → HPO phenotype	source = HPO / clinical presentation; high for PDAC, medium for related diseases
PARTICIPATES_IN	gene → Reactome pathway	source = Reactome stable ID; confidence = curated
TARGETS	drug → gene	mechanism, regulatory status, source (FDA label or trial); high if approved, medium if investigational
OCCURS_IN	variant → gene	source = PGS000725 lead SNP, reported/nearest gene; confidence = low

Table 1. Relationship types and the evidence carried as edge properties.

Confidence was assigned by rule, not by impression: a Module 2 edge is *high* only if RNA and protein change in the same direction and both have $p < 10^{-3}$; a driver edge is high if the gene is mutated in at least 15% of tumours; a GWAS edge is high only at genome-wide significance ($p < 5\times10^{-8}$), which leaves CLPTM1L ($p = 3.7\times10^{-7}$ in the original PanScan report) at medium. Druggability was a transparent tier from the drug layer (1.0 approved in PDAC, 0.8 approved elsewhere, 0.5 investigational, 0.2 none) because Module 3 tractability values were not carried into this graph.

2. Querying the graph

The two-hop query (Drug→Gene→PDAC) returned eight drug–gene pairs: **daraxonrasib** and sotorasib on KRAS, **erlotinib** and afatinib on EGFR, afatinib and neratinib on ERBB4, ensituximab on MUC5AC, and eprenetapopt on TP53. Only two are approved for PDAC: erlotinib with gemcitabine, and daraxonrasib, the first RAS inhibitor approved for pancreatic cancer (FDA, 26 August 2026). The others are approved elsewhere (sotorasib only for KRAS G12C, a rare PDAC allele) or failed in PDAC (ensituximab did not add benefit to second-line gemcitabine/nab-paclitaxel in a randomized phase 2). The five-hop shared-symptom query returned four candidates: pemigatinib, futibatinib and

ivosidenib via cholangiocarcinoma (shared: jaundice, weight loss), and ivacaftor via chronic pancreatitis (shared: abdominal pain, steatorrhea, diabetes mellitus). One full path reads:

```
pemigatinib —[TARGETS]→ FGFR2 —[ASSOCIATED_WITH]→ Cholangiocarcinoma —[HAS_PHENOTYPE]→ Jaundice
                                                                      ←[HAS_PHENOTYPE]— PDAC
```

3. Top targets by centrality, and one community

PageRank in the directed graph turned out to be uninformative for genes: every gene edge points out of the gene (to a disease or pathway), so directed PageRank mainly rewards nodes that receive edges and gives a gene extra weight only when a drug or variant points at it. I therefore fused undirected PageRank (overall connectivity) with gene evidence, fused = 0.5·percentile(PageRank) + 0.3·percentile(evidence) + 0.2·druggability, where evidence is the Module 2 score or −log10 GWAS p, ranked within its own type so the two scales are never compared directly.

Rank	Gene	Group	PageRank (undir.)	Degree	Module 2 rank / GWAS p	Fused score
1	ERBB4	multi-omics top 15	0.024	4	4	0.863
2	KRAS	driver	0.030	5	279	0.726
3	EGFR	driver / drug anchor	0.034	6	2444	0.715
4	MUC5AC	multi-omics top 15	0.019	3	10	0.679
5	FGFR2	related disease (CCA)	0.033	5	—	0.643
6	NOC2L	GWAS	0.014	2	8.4×10^{-14}	0.624
7	CEL	multi-omics top 15	0.019	3	11	0.587
8	NR5A2	GWAS	0.014	2	2.5×10^{-10}	0.564
9	TP53	driver	0.020	3	913	0.561
10	IDH1	related disease (CCA)	0.015	2	—	0.539

Table 2. Top 10 of 29 candidate biomarkers (full table: biomarkers_ranked.csv).

The ranking partly matches expected biology. **ERBB4** ranks first, consistent with Module 4, where it was the most druggable target (kinase-domain RMSD 2.82 Å against PDB 2R4B and a pocket overlapping the inhibitor site); it combines a top-5 multi-omics score with two approved pan-ERBB inhibitors and a pathway shared with EGFR. **KRAS** at rank 2 is exactly what PDAC biology predicts, and here the graph corrects Module 2, where KRAS ranked only 279 because its alteration is a sequence change rather than an expression change. Two hits are non-obvious and should be read as artefacts of the schema. **EGFR** ranks third although its Module 2 rank is 2444 and its protein barely changes; it is central because it carries two drugs and three pathways, i.e. it is well studied, not PDAC-specific. **FGFR2** is not a PDAC gene in my data at all; it enters through cholangiocarcinoma and ranks fifth only because it bridges the RAF/MAP kinase pathway, two FGFR drugs and a related disease. Centrality in a curated graph measures how much is known about a gene as much as how important it is. Conversely, the Module 2 top hits COL11A1, MUC5B, TNS4 and PLIN4 fall to ranks 13–24 because they have no drug and few pathway edges; TNS4 (Module 2 rank 3) is the clearest under-annotated candidate the graph cannot yet reward.

Greedy modularity found 12 communities. The five GWAS gene–variant pairs form isolated two-node communities, which shows the germline layer is not yet integrated with the rest of the graph. I interpret **community 2: CEL, CFTR, PRSS1, chronic pancreatitis, abdominal pain, steatorrhea, diabetes mellitus, digestion of dietary lipid and ivacaftor**. This is an exocrine-injury module: three acinar or ductal secretory genes whose germline variants cause pancreatitis, grouped with the symptoms of exocrine and endocrine failure. CEL is the bridge to PDAC. Its strong down-regulation in PDAC (RNA −6.1 log2, protein −2.7) was flagged in Module 2 as largely reflecting loss of normal acinar tissue, and here it sits with the pancreatitis genes, not the tumour-cell genes. The module therefore suggests a *pancreatitis-associated* PDAC subgroup: chronic pancreatitis is an established PDAC risk factor, and patients carrying CEL-HYB, PRSS1 or CFTR risk variants could be a genetically defined stratum for surveillance. By contrast, community 1 (KRAS, FGFR2, RAF/MAP kinase, EGFR signalling, four RAS/FGFR drugs) is the signalling module that supports the hypothesis below.

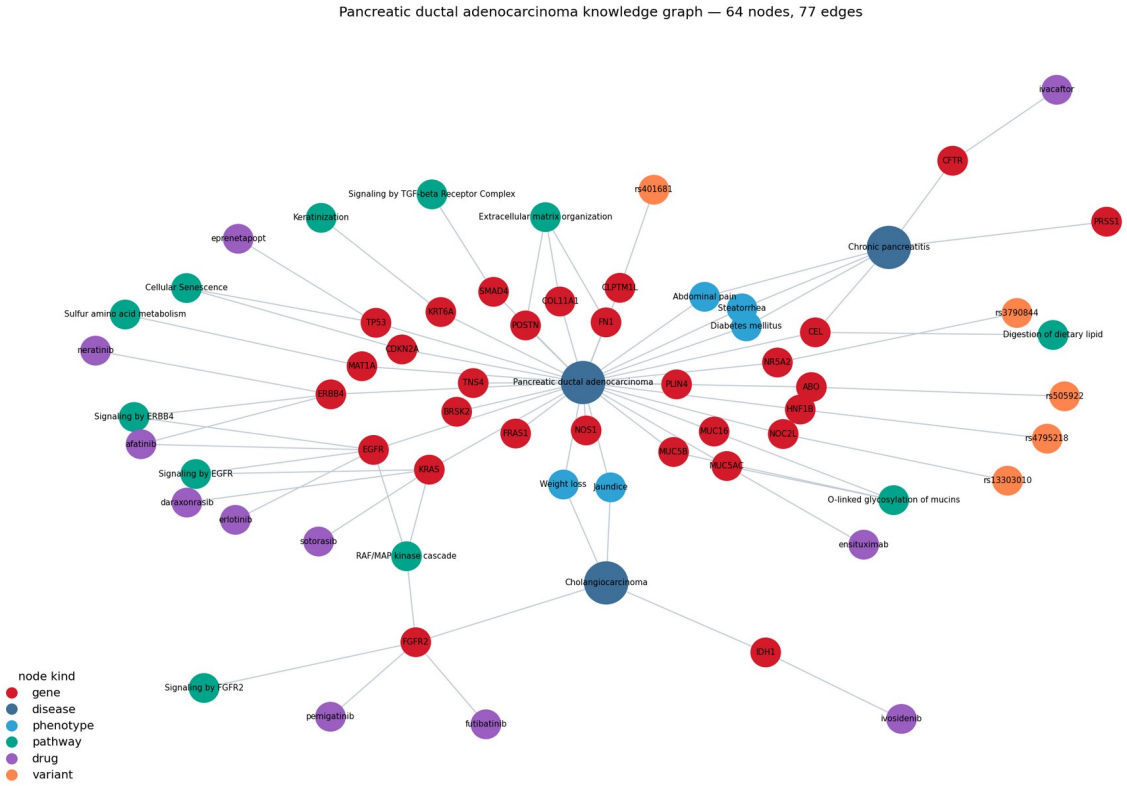


Figure 1. The PDAC knowledge graph (Kamada–Kawai layout), coloured by node kind; disease nodes are enlarged.

4. A hypothesis from a path

The path pemigatinib → FGFR2 → cholangiocarcinoma → jaundice → PDAC, read literally, is weak: a shared symptom says only that both tumours block the bile duct. Its value is that it points to a mechanism the graph also encodes. FGFR2 and KRAS fall in the same community through the RAF/MAP kinase cascade, so FGFR2 fusions can drive the same pathway that mutant KRAS drives. My hypothesis is that **FGFR inhibitors (pemigatinib, futibatinib) are a repurposing option for the KRAS-wild-type PDAC subset that carries FGFR2 fusions**, where FGFR2 replaces KRAS as the MAPK driver. This is plausible in the literature: about 10.7% of PDAC is KRAS-wild-type, and FGFR2 fusions were found in 5.2% of KRAS-wild-type tumours in a large Caris profiling cohort. It would also predict that daraxonrasib, which needs active RAS, works less well in these patients. To test it: (1) query CPTAC-PDAC (Module 2 data) for KRAS-wild-type tumours and check FGFR2 fusion calls and RNA/protein expression; (2) treat patient-derived organoids or cell lines with and without FGFR2 fusions using pemigatinib, measuring phospho-ERK and viability; (3) clinically, enrol FGFR2-fusion PDAC in tumour-agnostic FGFR-inhibitor basket trials and compare response with the cholangiocarcinoma benchmark.

5. An edge is evidence, not truth

The weakest edges are the **OCCURS_IN** edges. A GWAS lead SNP is usually non-coding, so linking rs505922 to ABO or rs13303010 to NOC2L uses the reported or nearest gene; of these, the source paper offers eQTL support only for NOC2L. With fine-mapping and eQTL or chromatin data, some of these edges would move to different genes, and the isolated GWAS communities might join the main graph or disappear. **HAS_PHENOTYPE** edges for related diseases are non-specific: jaundice is shared by any biliary obstruction, so the five-hop query would return drugs for gallstone-related or ampullary disease just as readily. Weighting phenotype edges by frequency (HPO annotations give "very frequent" vs "occasional") or symptom specificity would down-rank most of these paths. **TARGETS** edges ignore context: sotorasib targets only KRAS G12C, a rare PDAC allele, and afatinib's ERBB4 edge rests on biochemical potency rather than clinical activity in PDAC. Allele-level drug edges would remove sotorasib from the PDAC answer. Two **ASSOCIATED_WITH** edges are fragile: ERBB4 is RNA-protein discordant (RNA down, protein up), and CEL's association mostly reflects tissue composition. Finally, the centrality ranking is partly circular, because druggability and curated drug edges both reward well-studied genes. With an unbiased edge source, such as Open Targets association scores or

STRING protein interactions in place of hand curation, KRAS and the mucins would likely rise and EGFR and FGFR2 would fall.

Use of AI. I used Claude to write the code for the TODO cells, the identifier harmonization, the edge-evidence rules and the fused ranking, and to propose candidate HPO terms, Reactome pathways, drugs and GWAS SNPs. Nothing was accepted on its word. Each rsID and p-value was checked against the original GWAS papers (Klein 2018; Petersen 2010; Amundadottir 2009), and the notebook confirmed all five rsIDs are present in my Module 6 PGS000725 score file; each drug's target and approval status against FDA announcements and trial reports, and HPO and Reactome IDs against the HPO and Reactome sites. Inside the notebook, Ensembl REST confirms one ID per gene symbol and Reactome's ContentService confirms each pathway ID returns the expected name, so a wrong identifier fails visibly when the notebook runs rather than silently breaking a query. Every Cypher query was also cross-checked by recomputing its result in NetworkX on the same node and edge lists.

References

Klein AP et al. Genome-wide meta-analysis identifies five new susceptibility loci for pancreatic cancer. Nat Commun 9:556 (2018).

Petersen GM et al. A GWAS identifies pancreatic cancer susceptibility loci on chromosomes 13q22.1, 1q32.1 and 5p15.33. Nat Genet 42:224 (2010).

Amundadottir L et al. GWAS identifies variants in the ABO locus associated with susceptibility to pancreatic cancer. Nat Genet 41:986 (2009).

Jia G et al. Evaluating the utility of polygenic risk scores in identifying high-risk individuals for eight common cancers. JNCI Cancer Spectr 4:pkaa021 (2020) [PGS000725].

Fjeld K et al. A recombined allele of the lipase gene CEL and its pseudogene CELP confers susceptibility to chronic pancreatitis. Nat Genet 47:518 (2015).

U.S. FDA. FDA approves daraxonrasib for metastatic pancreatic adenocarcinoma (26 Aug 2026).

Solca F et al. Target binding properties and cellular activity of afatinib, an irreversible ErbB family blocker. J Pharmacol Exp Ther (2012).

Caris Life Sciences. Molecular characterization of KRAS wild-type tumors in patients with pancreatic adenocarcinoma.

Randomized phase 2 trial of NPC-1C (ensituximab) with second-line gemcitabine/nab-paclitaxel in advanced PDAC (NCT01834235). JAMA Netw Open.